TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	85.5 ± 1.75
Correlation Stability	89.2 ± 0.60
Distribution Consistency	88.3 ± 2.57
Entropy Preservation	87.3 ± 1.35
Mean Z Score Drift	0.1 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	99.4 ± 0.66
Feature Accuracy	96.3 ± 0.46
Novelty Score	0.9 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: hepatitis.csv

1. Dataset Portrait

Attribute	Value
Rows	155
Columns	20
Numeric features	20
Categorical features	0
Missing cells (%)	5.4%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	41.200	41.177	-0.023	0.1%	OK
Sex	1.103	1.169	+0.066	6.0%	OK
Steroid	1.506	1.444	-0.063	4.2%	OK
Antivirals	1.845	1.839	-0.006	0.3%	OK
Fatigue	1.351	1.331	-0.020	1.5%	OK
Malaise	1.604	1.565	-0.039	2.5%	OK
Anorexia	1.792	1.815	+0.022	1.2%	OK
Liver Big	1.828	1.847	+0.019	1.0%	OK
Liver Firm	1.583	1.661	+0.078	4.9%	OK
Spleen Palpable	1.800	1.831	+0.031	1.7%	OK
Spiders	1.660	1.702	+0.042	2.5%	OK
Ascites	1.867	1.887	+0.020	1.1%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	85.5 ± 1.75
Correlation Stability (*2)	89.2 ± 0.60
Distribution Consistency (*3)	88.3 ± 2.57
Entropy Preservation (*4)	87.3 ± 1.35
Mean Z Score Drift (*5)	0.1 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	99.4 ± 0.66
Feature Accuracy (*10)	96.3 ± 0.46
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8064516129032258	0.8064516129032258	0.8064516129032258	0	
F1	N/A	0.0	N/A	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.8516129032258064	0.8516129032258064	0.012903225806451613	
F1	0.5718392134181608	0.5011478717361071	0.510107800696036	-0.06173141272212479	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8741935483870968	0.8516129032258064	0.864516129032258	-0.009677419354838724	
F1	0.625869620575503	0.3788888888888889	0.5923465423465425	-0.03352307822896058	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7806451612903226	0.7935483870967742	0.7903225806451613	0.009677419354838712	
F1	N/A	0.02	N/A	0	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7838709677419355	0.8	0.8419354838709676	0.05806451612903224	
F1	0.33387339784398606	0.344017094017094	0.5242268842268843	0.19035348638289817	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7935483870967742				

2

0.774193548387096

8

0.806451612903225

8

0.012903225806451612

3

F1	0.44524475524475526	0.38597402597402597	0.42265900765900766	-0.022585747585747606	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8516129032258064	0.829032258064516	0.8354838709677418	-0.016129032258064523	
F1	0.5894355644355643	0.4685858585858555	0.5328937728937728	-0.05654179154179154	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7903225806451613	0.8161290322580644	0.8516129032258064	0.06129032258064515	
F1	0.561449871078354	0.5901174504889675	0.6641604010025063	0.1027105299241522	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.8419354838709678	0.8419354838709676	0.003225806451612915	
F1	0.5132478632478632	0.4884163059163059	0.5216483516483517	0.008400488400488404	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8516129032258064	0.8193548387096774	0.8483870967741935	-0.003225806451612915	
F1	0.5385593354014405	0.421430203783145	0.5459607059607061	0.007401370559265296	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
2	50.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0	2.0
3	78.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0
5	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
8	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
9	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
10	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
11	39.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0
12	32.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
13	41.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
14	30.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
15	47.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
16	38.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
17	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
18	40.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
19	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
20	38.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	36.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
2	39.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0	1.0
3	40.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0
4	20.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
5	30.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0
6	18.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	61.0	1.0	1.0	2.0	1.0	1.0	2.0	nan	nan	2.0
8	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
9	30.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
10	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	34.0	2.0	1.0	2.0	1.0	1.0	1.0	1.0	1.0	1.0
2	51.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	32.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0
4	13.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
5	69.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0
6	43.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0	2.0
7	60.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	1.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
9	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	36.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	76.0	1.0	1.0	2.0	1.0	2.0	1.0	1.0	1.0	1.0
2	54.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
3	35.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	18.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
5	35.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	44.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0
7	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	42.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	50.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
2	4.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
3	45.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0
4	52.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0
5	41.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0
6	68.0	2.0	1.0	2.0	1.0	1.0	1.0	1.0	1.0	2.0
7	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
9	25.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
10	38.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	27.0	1.0	1.0	2.0	2.0	1.0	2.0	2.0	1.0	2.0
2	29.0	1.0	2.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0
3	35.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0
4	37.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0
5	60.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0	2.0
6	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	30.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	45.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	36.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	45.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
2	46.0	1.0	1.0	2.0	1.0	1.0	1.0	1.0	1.0	2.0
3	54.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
5	47.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
6	32.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
8	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
9	51.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
10	40.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	45.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
2	40.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0
3	40.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	47.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0
5	41.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
6	44.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
7	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
9	30.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	59.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
2	36.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0
3	44.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
4	29.0	1.0	2.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
5	45.0	2.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
6	36.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
7	65.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
8	36.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
9	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
10	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	35.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0
2	51.0	2.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
3	70.0	1.0	2.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0
4	54.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
5	41.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	33.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
7	38.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	44.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
9	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	28.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	29.0	1.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0
2	42.0	1.0	2.0	2.0	2.0	1.0	2.0	2.0	1.0	1.0
3	47.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	36.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
5	37.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0
6	40.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
7	56.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	28.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	49.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0